

Revised Generation Interconnection System Impact Study Report

For Queue Project Q13-0292

Church Rock 138 kV
August 22, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0292. The Interconnection Customer proposes a 389.0 MW wind facility interconnecting at the Church Rock 138 kV substation in the West Virginia area. The request was received on November 20, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 168 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$21,527,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Balfour / Preston North 138 kV	International Paper / Flatlick 500 kV	118.9	270	43.2
Balfour / Church Rock Substation 138 kV L	Flatlick / International Paper 500 kV	118.2	277	43.2
Preston N / Myrtle-Date 138 kV	Flatlick / International Paper 500 kV line	117.7	268	43.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Balfour / Preston North 138 kV	Rebuild line with higher-capacity conductor	\$13,850,000	96% (\$13,255,000)
Balfour / Church Rock Substation 138 kV L	Rebuild line with higher-capacity conductor	\$4,655,000	96% (\$4,455,000)
Preston N / Myrtle-Date 138 kV	Rebuild line with higher-capacity conductor	\$3,988,000	96% (\$3,817,000)
	Total allocated to project		\$21,527,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.